

Calibration decisions after lunch

September 23, 2026 | National housing geography adopted for documents; other target definitions and weights remain open

Keep the 2007 stationary reference and most of the existing target construction. The completed review does not justify rebuilding the July strategy. The main first-birth housing comparison does need a common measurement rule before it can support an adopted calibration.

Decisions on fertility, wealth and housing levels can be made now. The first-birth diagnostic is a separate calculation from saved policies and must not hold up those decisions. Confidence flags below assess measurement comparability, not how well a selected model fits.

The present reference combines 2004/2006 CPS fertility stocks, 2003-2006 natality timing, 2003/2005 PSID wealth and national 2005/2006 ACS housing. These are nearby-year measurements used for a stationary 2007 economy, not one identical population observed in 2007.

Six author decisions

Reference economy and housing population

Recommendation: The author has selected the current 2007 reference with national US housing targets. The four national housing values replace the prior 42-metro document values for the next comparison.

Next action: Defer loss calculation and rescoring until the next run. Record the complete national target/weight contract before that comparison.

Fertility stocks, timing and normalization

Recommendation: Keep both CPS shares, both grouped NCHS timing targets and the separately imposed 2.1 normalization. Use one stated within-cell timing convention; retain the implemented uniform-time approximation for now.

Next action: Write the accepted population, age and timing conventions into the contract. Recompute only if a different convention is chosen.

Saving and bequest targets

Recommendation: Retain the three signed-off economic objects: aggregate wealth/gross earnings, annual bequests/wealth and living-old wealth-to-income dispersion. Keep the existing scalar values for this comparison.

Next action: No new data audit is needed to retain these rows. A chosen levels-dispersion alternative can be rescored on saved model values under a separately named contract.

Children in the family-room and recent-parent rows

Recommendation: Retain the current selected national resident-own-child family-room row and the national recent-parent ownership row, with the existing stock-versus-flow qualification for recent parents. The minor-child definition remains a proposed choice pending discussion.

Next action: Defer loss calculation and rescoring until the next run. An exact recent-child-age comparison remains separate saved-policy measurement work.

First-birth housing response

Recommendation: Keep the original PSID regression as empirical evidence; require an explicitly matched auxiliary statistic on data and simulated model histories before adopting this row as evidence for one utility specification.

Next action: Declare birth/interview timing, sample, joint observation histories, controls and weights on both sides before calculating the candidate fit. If education is absent from the model, any reduced empirical regression must drop it on the data side too and be labeled a new candidate. The lunch calculation only exposes origin and destination branch responses.

Weights and specification choice

Recommendation: Treat weighting and uncertainty as open decisions for the national population. The frozen weights remain historical comparison weights and are not adopted for the new population.

Next action: Defer loss calculation and rescoring until the next run. Revisit utility only after that comparison; new behavioral solutions are needed for a refit, not for a purely scalar target rescore.

Current target table

Parameter connections describe economic margins, not one-to-one identification. Confidence concerns measurement comparability, not model fit.

Target	Value	Parameter connection	Confidence / decision
Mean children ever born, ages 46+	2.1000	Fertility utility scale psi (separate normalization)	Confident as normalization: Keep; do not label it period TFR or a measured cohort target.
Childless women, ages 40-44	19.8279%	First-birth taste scale and birth cost jointly	Medium: Keep empirical share; accept stated female-exposure and age-cell approximation.
Exactly one child, among mothers 40-44	21.3655%	Continuation-birth taste scale; other fertility preferences	Medium: Keep conditional stock; it is not a second-birth hazard.
Mean first-birth age, grouped	25.9763 years	First-birth cost and taste scale jointly	Medium: Keep grouped data/model convention; no substitution of the raw mean.
Share of first births at ages 30+	24.9278%	First-birth cost and taste scale jointly	Medium: Keep; same period, support and timing contract as mean age.
Aggregate wealth / annual gross earnings	6.14586	Annual discount factor beta	Confident on audited definition: Retain the July stock/flow definition and current early-period value.
Annual bequest flow / wealth	0.8800%	Bequest strength theta0	Medium: Retain borrowed restriction; 5% tolerance is assigned, coverage approximate.
Old wealth/income, p90/p50	3.51594	Bequest wealth shift theta1; retirement saving	Medium: Retain signed-off object; pure wealth-level alternative is an explicit optional choice.
Mean rooms, capped at 9	5.60789 rooms	Housing supply intercept H0; demand jointly	Confident conditional on population: National US value selected; prior 42-metro value is historical.
Ownership, ages 30-55	67.6260%	Owner service multiplier chi; credit restrictions fixed	Confident conditional on population: National US value selected; no identified scalar or age-mask bug explains low model ownership.
First-birth room response, -1 to +3	0.720246 rooms	Parenthood housing floor h_P, or share-jump parameter	Problematic model/data match: Retain empirical evidence; repair common estimator before using fit for adoption.
Rooms, 3+ versus 1-2 resident children	0.385100 rooms	Housing preferences; child-share slope in alternative	Medium: National resident-own-child value selected; a national minor-child candidate is not yet measured.
Recent-parent ownership gap	12.7608 pp	Owner service multiplier and parenthood housing demand	Medium; recency approximation: National US value selected provisionally; exact child-age history measurement remains distinct work.

Decision evidence and implementation boundary

Reference economy and housing population

Evidence: The verified national values are mean rooms 5.607885960068579, ownership 0.6762604168538028, family rooms 0.38509964969278165 and recent-parent ownership gap 0.12760836356692162.

Alternative: The former 42-metro document values remain historical comparison values; they are not the current selected housing population.

Consequence: The selected national ownership target is higher than the former 42-metro target. This decision does not by itself explain model fit.

Fertility stocks, timing and normalization

Evidence: The empirical scalars and implemented age mapping reproduce. Mean first-birth age 25.9763 is the data mapped to model age cells, not an arithmetic error. Raw recorded-age mean 25.1608 is a different object. The 2.1 anchor concerns age-46+ mean children ever born and is not a measured period TFR.

Alternative: Adopt a different birth-time or maternal-age-support convention jointly for data and model. Do not replace the grouped mean with the raw mean while leaving its model observer unchanged.

Consequence: The model approximates one reproductive household member and within-cell age exposure. Accepting that approximation permits decisions now; it does not certify exact female survey exposure.

Saving and bequest targets

Evidence: The July review explicitly selected these objects. Current arithmetic reproduces. The bequest weight is a chosen 5% tolerance, not a sampling standard error. Constant modeled pension income makes the model's old-age wealth and wealth/income percentile ratios coincide.

Alternative: If the intended third object is pure wealth dispersion, choose the already measured same-sample levels ratio 4.919429 (SE 0.579804), replacing the current wealth/income ratio 3.515935 (SE 0.306911). This is a conceptual choice, not a mandatory code correction.

Consequence: Retaining the current dispersion accepts omitted retirement-income heterogeneity as a measurement approximation. Neither option fixes or validates the entrant-wealth mapping.

Children in the family-room and recent-parent rows

Evidence: The national family-room target is 0.38509964969278165 and the national recent-parent ownership gap is 0.12760836356692162. The linked-minor 0.336220 candidate was measured for 42 metros, not nationally.

Alternative: A national linked-minor candidate has not yet been measured. The existing 42-metro minor-child candidate is not a national alternative.

Consequence: The population choice does not silently alter the child grouping. Minor age and stochastic model dependence are still not exactly the same object.

First-birth housing response

Evidence: The 0.720246 target and clustered SE 0.085260 reproduce exactly. The current scored model contrast instead compares two branches from the same pre-birth households at the next model date, permitting another birth only in the treated branch. These are different estimators.

Alternative: Explicitly accept the current forced-branch comparison as a provisional approximation. It must not be described as the identical PSID regression or used to claim that a housing mechanism is rejected by the data.

Consequence: A longer empirical window, division by two, or a favorable timing projection is not an automatic correction. Preserve this identifying information for the housing-preference block; do not drop the row simply because it is difficult to fit.

Weights and specification choice

Evidence: Scales mix empirical sampling uncertainty, annual temporal variation and an external bequest tolerance; cross-moment covariance is not used. The floor estimates nine structural coordinates, the shares alternative ten, with fertility utility normalized separately.

Alternative: Choose a weighting rule on substantive grounds, name the full replacement contract and preserve all affected identifying information. A new rule is not a revision of old scores.

Consequence: A lower loss is useful search evidence but not an efficient-SMM test, identification proof or numerical-convergence certificate.

Other documented choices

Entry wealth: retain heterogeneity and the reviewed 18-24 childless-renter proxy. The running inherited law and the new common-income-scale candidate are different laws. The candidate has positive current renter slack, but no full continuation/equilibrium validation or adoption. Choose units explicitly before a new refit; zero entry is not the baseline recommendation.

External housing depreciation: the inspected serialized input uses 1.1% annually; the slides say 2%. Existing comparisons use 1.1%. Reconcile this restriction before a new refit; a change would require new solutions, not target rescoring. No silent model change is made here.

Parameter connections in the table describe informative economic margins, not one-to-one identification. Fixed borrowing, earnings and housing restrictions affect many of these moments.

Scope and evidence

lunch_decision_evidence.json
target_decisions_20260923.csv
july_decision_review.json
overnight/phase5_lead_review.json
overnight/phase6_lead_review.json
overnight/report_tables/target_contract.csv
lunch_first_birth_empirical_inventory.json
lunch_first_birth_model_inventory.json
lunch_first_birth_comparison.csv
lunch_first_birth_helper_review.json
lunch_first_birth_final_review.json
lunch_targets_and_selected_moments.csv

Not done or claimed

National housing geography and four corresponding values adopted for documents; other target definitions and weights remain open.

No new target-weight contract, target removal or utility adoption.

No matched simulated-panel event-study estimate yet.

No household or equilibrium solve, calibration search, author-owned manuscript edit or new fit.

Supplemental first-birth diagnostic

Same-origin successful-first-birth households, cloned into a birth branch and a childless control; stationary saved policy at origin and four years later. No empirical regression is simulated.

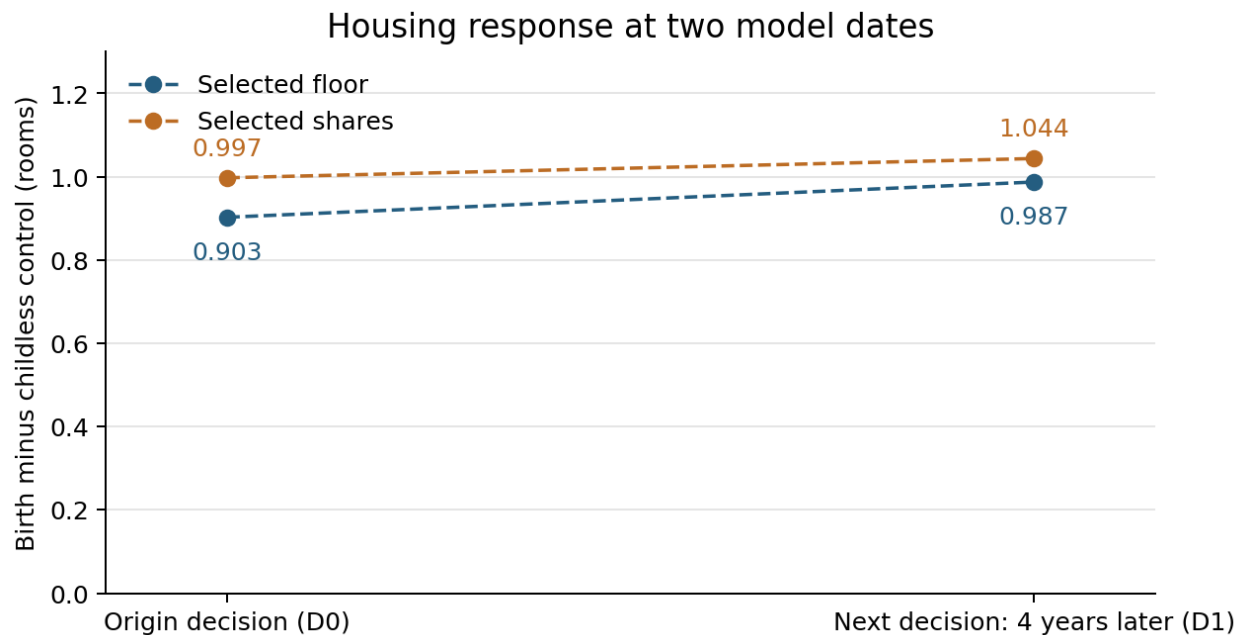
The immediate response is already 0.903 rooms under the selected floor and 0.997 under selected child-dependent shares. Four years later the corresponding responses are 0.987 and 1.044. Moving the native reading from the destination date to the origin reduces the response by only 0.085 and 0.046 rooms, respectively.

These calculations establish that most of this model branch response occurs at the birth decision. They do not establish that the empirical miss survives a common estimator: the PSID coefficient contrast uses different comparison households, timing, controls and weights. Neither native date is automatically the empirical -1/+3 contrast.

Both cases use persistent earnings with 15 income states and inherited heterogeneous entry wealth, but are separately recalibrated selected points. Their difference cannot be attributed to utility alone. Each destination response reproduces its original scored value exactly; the full contracts and checkpoint identities were checked. No new household or equilibrium was solved.

The returned mass checks pass unchanged. The floor has zero destination projection mass; the shares case has tiny recorded projections around 1.55e-167, not exactly zero. This measurement replay adds no grid-convergence or continuation-value certificate.

Statistic (rooms)	PSID	Floor	Shares
Original empirical -1/+3 contrast	0.720246 (SE 0.085260)	Different estimator	Different estimator
Immediate native response D0	Not computed	0.902597	0.997423
Four-year destination response D1	Not computed	0.987251	1.043655
Additional native response D1-D0	Not computed	0.084654	0.046232
Gap from saved scored D1	Not applicable	0 exactly	0 exactly



Supplemental diagnostic only. Dashed segments connect two model dates; no annual path or birth-date interpolation is assumed. This is not an overlay of the PSID event-study coefficients. The standard 17 plots per selected case remain unchanged.